



MENAI HIGH SCHOOL

ASSESSMENT TASK NOTIFICATION



Computing Technology - Assessment Task 2

Teacher: Ms A Kasper

Due Date: Term 4, Week 1

Task Description

In this project, students will work in a small team to design, build, and program a Micro:bit Sumo Robot that competes in a class arena challenge. Your robot must be able to:

- Follow a line to enter the arena
- Detect and push an opponent robot out of the ring
- (Optional for larger groups) Detect and navigate around an obstacle on the entry line

As part of the project, you will create design drawings, a circuit diagram, and working code for your robot. You will also test and evaluate your robot, document your algorithms and debugging process, and reflect on the broader impacts of robotics and automation in society.

The following points must be completed within your submission:

Identifying and Defining:

- **Project Purpose:** In your own words, describe the purpose of the Micro:bit Sumo Robot Challenge. Why are we building these robots? What will success look like? (CT5-EVL-01)
- **Team Roles:** Define the role of each team member (line follower, sumo attacker, obstacle navigation). Write down who is responsible for which part of the project. (CT5-COL-01)

Researching and Planning:

- **Design drawings:** Create annotated sketches or CAD models of your robot (CT5-COL-01).
- **Identify and explain components:** Explain the use of all sensors and actuators that you will need. Identify the uses of these and provide examples of how they can be used in mechatronics (CT5-THI-01).
- **Algorithms:** Write flowcharts for your robot's behaviours (CT5-OPL-01).

- **Circuit diagram:** Draw a labelled circuit diagram showing inputs (sensors), processes (micro:bit), and outputs (motors) (CT5-COM-01).
- **Research:** Investigate how mechatronic systems have evolved in the real world, and consider their social and ethical impacts (CT5-SAF-01, CT5-EVL-01).

Producing and Implementing:

- **Code development:** Write modular, structured code for your robot using loops, functions, and conditionals (CT5-DPM-01, CT5-OPL-01). Ensure that code is well-commented and readable.
- **Build and integrate:** Construct the robot and connect hardware components safely and securely (CT5-SAF-01).
- **Practical challenge:** Test your robot in the arena to:
Follow the entry line
Push an opponent robot
Navigate around an obstacle (For groups of 3) (CT5-DPM-01)
- **Collaboration:** Demonstrate effective teamwork and shared decision-making (CT5-COL-01).

Testing and Evaluating:

- **Test cases:** Create a table of inputs (e.g. “line detected”), expected outputs (e.g. “turn right”), and actual outputs (CT5-DAT-02).
- **Debugging:** Show evidence of how you identified and fixed errors in your code (CT5-DPM-01).
- **Evaluation:** Reflect on how well your robot met the requirements, and suggest improvements (CT5-EVL-01).
- **Reflection:** Discuss the broader impacts of robotics and automation on different communities, including accessibility, ethics, and data use (CT5-SAF-01, CT5-DAT-01, CT5-COM-01)

Final Demonstration

At the end of the project, all groups will compete in a class Sumo Robot Competition. This will be your chance to showcase your robot’s design, coding, and teamwork as it performs against other teams’ robots in the arena.

** To ensure that you get the best results possible, make sure that you are completing all

areas of the assessment matrix **

Submission Details:

- The task will be completed both in class and at home.
- All word-processed files must be saved as a pdf and all computer files must be compressed before uploading it to Google Classroom by the submission time on the due date. An E-Grade will be awarded for late submission.

** Please note that where plagiarism has occurred, students will receive an E-Grade automatically and be asked to complete the task again. **

Outcomes Assessed

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CT5-SAF-01	Selects and applies safe, secure, and responsible practices in the ethical use of data and computing technology
CT5-DPM-01	Applies iterative processes to define problems and plan, design, develop and evaluate computing solutions
CT5-DAT-01	Explains how data is stored, transmitted, and secured in digital systems and how information is communicated in a range of contexts
CT5-COM-01	Communicates ideas, processes and solutions using appropriate media
CT5-OPL-01	Designs, produces, and evaluates algorithms and implements them in a general-purpose and/or object-oriented programming language
CT5-THI-01	Applies computational, design and systems thinking to the development of computing solutions
CT5-DAT-02	Acquires, represents, analyses, and visualises simple and structured data
CT5-DES-01	Designs and creates user interfaces and the user experience

Assessment Task 2: Marking Matrix					
	A	B	C	D	E
Select a topic and formulate a research question or hypothesis (Outcome CT5-DAT-02)	Formulates a highly relevant and insightful research question or hypothesis. Demonstrates deep understanding of the topic.	Constructs a relevant research question or hypothesis. Shows good understanding of the topic.	Develops a basic research question or hypothesis that is somewhat relevant. Shows basic understanding of the topic.	Identifies a limited research question or hypothesis. Shows minimal understanding of the topic.	States a vague or irrelevant research question or hypothesis. Shows little to no understanding of the topic.
Design and document a method for collecting the required data (Outcome CT5-DAT-02)	Designs and documents a comprehensive and innovative method for data collection. Clearly outlines detailed steps.	Constructs and documents a clear and effective method for data collection. Outlines steps adequately.	Develops and documents a basic method for data collection. Provides some steps but lacks detail.	Identifies and documents a limited method for data collection. Steps are unclear or incomplete.	States and documents an inadequate or unclear method for data collection. Steps are missing or poorly described.
Discuss how you will ensure the data is collected and used ethically and responsibly (Outcome CT5-SAF-01)	Presents clear and comprehensive evidence of data collection. Discusses in-depth ethical considerations and provides a comprehensive plan for responsible data collection and use.	Shows adequate evidence of data collection. Explains ethical considerations and outlines a clear plan for responsible data collection and use.	Provides basic evidence of data collection. Describes basic ethical considerations and provides a basic plan for responsible data collection and use.	Demonstrates limited evidence of data collection. Identifies limited ethical considerations with an incomplete or unclear plan for responsible data collection and use.	Offers minimal or no evidence of data collection. Mentions minimal or no ethical considerations with no clear plan for responsible data collection and use.
Research different types of infographics and evaluate the best approach for displaying the collected data. (Outcome CT5-DPM-01)	Evaluates various types of infographics critically and selects the best approach with insightful justification.	Compares different types of infographics and selects an appropriate approach with clear justification.	Describes several types of infographics and chooses a basic approach with some justification.	Identifies a few types of infographics and selects a limited approach with minimal justification.	Mentions one or two types of infographics with little or no justification for the selected approach.
Create a storyboard or wireframe of the infographic, annotating the layout and types of visualisations (Outcome CT5-DES-01)	Creates a highly detailed and well-annotated storyboard/wireframe. Clearly indicates layout and visualisations with insightful annotations.	Develops a detailed and annotated storyboard/wireframe. Clearly indicates layout and visualisations with appropriate annotations.	Produces a basic storyboard/wireframe with some annotations. Shows layout and visualisations with basic annotations	Attempts a limited storyboard/wireframe with minimal annotations. Layout and visualisations are unclear or incomplete.	Provides a vague or incomplete storyboard/wireframe with few or no annotations. Layout and visualisations are poorly indicated.
Design the user interface (UI) and user experience (UX) using design principles to ensure the	Designs an exceptional UI/UX that is highly user-friendly and visually appealing, applying	Constructs a user-friendly and visually appealing UI/UX, applying design principles effectively.	Develops a basic UI/UX that is generally user-friendly and visually appealing, applying some	Creates a limited UI/UX with minimal user-friendliness or visual appeal, applying few	Produces an inadequate UI/UX that is neither user-friendly nor visually appealing,

infographic is user-friendly and visually appealing. (Outcome CT5-DES-01)	advanced design principles. Examples: harmonious color schemes, readable and aesthetically pleasing fonts, well-structured layout, intuitive navigation.	Examples: good color contrast, clear and consistent typography, logical layout, intuitive navigation.	design principles. Examples: basic color contrast, standard readable typography, simple layout, basic navigation.	design principles. Examples: inconsistent colors, difficult-to-read typography, cluttered layout, confusing navigation.	with little to no application of design principles. Examples: clashing colors, unreadable fonts, chaotic layout, difficult navigation.
Implement the designs and algorithms to create the interactive elements of the infographic. (Outcome CT5-OPL-01)	Implements the designs and algorithms flawlessly, creating highly effective interactive elements. Examples: dynamic charts that update in real-time, interactive maps with tooltips, and filterable data views.	Develops the designs and algorithms effectively, creating functional interactive elements. Examples: interactive charts with basic tooltips, clickable elements that navigate to different sections, and sortable tables.	Creates the designs and algorithms with some effectiveness, resulting in basic interactive elements. Examples: static charts with limited interactivity, simple tooltips, and basic navigation elements.	Attempts the designs and algorithms with limited success, resulting in partially functional interactive elements. Examples: interactive elements that work inconsistently, tooltips that display incorrect information, or navigation elements that are hard to use.	Produces the designs and algorithms with minimal effectiveness, resulting in non-functional or ineffective interactive elements. Examples: broken charts, tooltips that do not appear, and navigation elements that do not work.
Acquire, clean, and pre-process the dataset to ensure accuracy and usability. (Outcome CT5-DAT-02)	Acquires, cleans, and preprocesses the dataset thoroughly, ensuring high accuracy and usability. Examples: removes all duplicate entries, corrects all errors, handles missing data effectively, normalises data formats, and validates data consistency.	Processes the dataset effectively, ensuring good accuracy and usability. Examples: removes most duplicate entries, corrects most errors, addresses most missing data, standardises data formats, and checks for data consistency.	Prepares the dataset adequately, ensuring basic accuracy and usability. Examples: removes some duplicate entries, corrects some errors, handles some missing data, standardises data formats to some extent, and performs basic consistency checks.	Handles the dataset with limited success, resulting in some inaccuracies or usability issues. Examples: misses many duplicate entries, leaves several errors uncorrected, inconsistently handles missing data, and performs minimal standardisation and consistency checks.	Attempts to process the dataset with minimal success, resulting in significant inaccuracies or usability issues. Examples: leaves most duplicate entries, corrects few errors, ignores most missing data, performs little to no standardisation, and does not check for consistency.
Use the selected software to generate visualizations and integrate them into the final interactive infographic (Outcome CT5-OPL-01)	Generates multiple highly effective visualisations (minimum of 3) using the selected software and integrates them seamlessly into the final interactive infographic.	Creates several effective visualisations (minimum of 3) using the selected software and incorporates them well into the final interactive infographic.	Develops basic visualisations (minimum of 3) using the selected software and includes them adequately in the final interactive infographic.	Produces limited visualisations (less than 3) using the selected software and integrates them with some issues into the final interactive infographic.	Attempts to create visualisations (less than 3) with minimal success using the selected software and adds them poorly to the final interactive infographic.
Conduct user testing to gather feedback on usability and effectiveness.	Conducts comprehensive user testing, gathering extensive feedback on	Performs thorough user testing, gathering substantial feedback on	Carries out basic user testing, gathering some feedback on usability and	Attempts limited user testing, gathering minimal feedback on usability and	Provides inadequate user testing, gathering little to no feedback on

<i>(Outcome CT5-DPM-01)</i>	usability and effectiveness.	usability and effectiveness.	effectiveness.	effectiveness.	usability and effectiveness.
Evaluate the infographic against the initial goals and make necessary improvements based on feedback. <i>(Outcome CT5-DPM-01)</i>	Evaluates the infographic comprehensively against the initial goals and implements necessary improvements thoroughly based on detailed feedback.	Assesses the infographic effectively against the initial goals and makes necessary improvements based on substantial feedback.	Reviews the infographic adequately against the initial goals and applies some improvements based on basic feedback.	Examines the infographic minimally against the initial goals and makes limited improvements based on minimal feedback.	Considers the infographic poorly against the initial goals and applies little to no improvements based on feedback.
Write a report reflecting on the iterative process, challenges faced, and how they were overcome. <i>(Outcome CT5-COM-01)</i>	Writes a detailed and insightful report reflecting on the iterative process, challenges faced, and solutions implemented effectively.	Produces a thorough report reflecting on the iterative process, challenges faced, and solutions applied.	Develops a basic report reflecting on the iterative process, challenges faced, and solutions attempted.	Creates a limited report with minimal reflection on the iterative process, challenges faced, and solutions considered.	Provides an inadequate report with little to no reflection on the iterative process, challenges faced, or solutions addressed.